Exercise 2.9

In Exercises 1 and 2, find the vector x determined by the given coordinate vector $[x]_B$ and the given basis B.

1. $B = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ -1 \end{bmatrix} \right\}, [x]_B = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

Solution: By Defn, $x = 3\begin{bmatrix} 1 \\ 1 \end{bmatrix} + 2\begin{bmatrix} 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$

In Exercises 3–6, the vector x is in a subspace H with a basis $B = \{b_1, b_2\}$. Find the B-coordinate vector of x.

3. $b_1 = \begin{bmatrix} 1 \\ -4 \end{bmatrix}, b_2 = \begin{bmatrix} -2 \\ 7 \end{bmatrix}, x = \begin{bmatrix} -3 \\ 7 \end{bmatrix}$

Solution: Let, c_1 and c_2 be two arbitrary reals.

By Defn. of Basis, $c_1b_1 + c_2b_2 \in H$. Let, $x = c_1b_1 + c_2b_2$.

$$\therefore \begin{bmatrix} -3 \\ 7 \end{bmatrix} = c_1 \begin{bmatrix} 1 \\ -4 \end{bmatrix} + c_2 \begin{bmatrix} -2 \\ 7 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} -3 \\ 7 \end{bmatrix} = \begin{bmatrix} 1 & -2 \\ -4 & 7 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} \quad \text{(By Defn. of } Ax\text{)}$$

By Theorem 3 from Chapter 1, the above matrix equation has the same solution set as the system of equations corresponding to the augmented matrix